Problem 1

Suppose that $A \in \mathbb{R}^{m \times n}$, $\mathbf{v} \in \mathbb{R}^n$ and $B \in \mathbb{R}^{n \times r}$. Determine the number of floating point operations (*ie* additions, multiplications and divisions) required to compute (a) $A\mathbf{v}$ and (b) AB.

Answer:

(a) $A^{m \times n} \times \mathbf{v}^{n \times 1} = \mathbf{b}^{m \times 1}$. This requires $m \times n$ operations. When doing matrix vector multiplication, you start by multiplying the first row of the matrix (n elements) with the column vector (n elements), which requires n operations. You have to repeat this process for all m rows of the matrix, hence this requires $m \times n$ floating point operations.

(b) $A^{m \times n} \times \ B^{n \times r} = C^{m \times r}$. This requires $(m \times r)n$ operations. The resulting matrix C has $m \times r$ elements and to get each of those, you need to multiply a row of A with a column of B, which is n floating point operations. Thus, there are a total of $m \times r \times n$ floating point operations.

Problem 2. (10 points)

Let $\mathbf{v} = [1, 0, 5, -3]^\top$ and $\mathbf{w} = [-2, 4, 5, 1]$.

1. Compute $\|\mathbf{v}\|_1$, $\|\mathbf{w}\|_3$ and $\|\mathbf{v}\|_\infty$.

2. Verify that $\langle \mathbf{v}, \mathbf{w} \rangle \leq \|\mathbf{v}\|_2 \|\mathbf{w}\|_2$ by computing both sides.

Answer:

(a) $\|\mathbf{v}\|_1 := \left(\sum_{i=1}^{4} |v_i|^1 \right)^{1/1}$
$= |1| + |0| + |5| + |-3| = 9$

$\|\mathbf{w}\|_3 := \left(\sum_{i=1}^{4} |w_i|^3 \right)^{1/3}$
$= (|-2|^3 + |4|^3 + |5|^3 + |1|^3 = 9)^{1/3}$
$= (8 + 64 + 125 + 1)^{1/3}$
$= \sqrt[3]{198} \approx 5.828$

$\|\mathbf{v}\|_\infty := max_i |v_i|$
when $i = 3$, $|v_i| = 5$, which is the maximum. Therefore $\|v_\infty\| = 5$

(b) $\langle \mathbf{v}, \mathbf{w} \rangle \leq \|\mathbf{v}\|_2 \|\mathbf{w}\|_2$

LHS $= \langle \mathbf{v}, \mathbf{w} \rangle = \langle [1, 0, 5, -3]^\top, [-2, 4, 5, 1]^\top \rangle = (1 \times -2) + (0 \times 4) + (5 \times 5) + (-3 \times 1) = 20$

RHS $= \|\mathbf{v}\|_2 \|\mathbf{w}\|_2 = \sqrt{1^2 + 0^2 + 5^2 + 3^2} \times \sqrt{2^2 + 4^2 + 5^2 + 1^2}$
$= \sqrt{35} \times \sqrt{46} = \sqrt{1610} \approx 40.125$

$20 \leq 40.125$
LHS $\leq$ RHS
$\langle \mathbf{v}, \mathbf{w} \rangle \leq \|\mathbf{v}\|_2 \|\mathbf{w}\|_2$ ■

Problem 3. (10 points)

Show that $\|\mathbf{v}\|_p := \left(\sum_{i=1}^{n} |v_i|^p\right)^{1/p}$ is a norm for $1 \leq p \leq \infty$. *Hint: you will need to use Hölder's inequality:*

$$\sum_{i=1} |v_i w_i| \leq \|\mathbf{v}\|_p \|\mathbf{w}\|_q \text{ for all } \mathbf{v}, \mathbf{w} \in \mathbb{R}^n \text{ and } q := 1 - \frac{1}{p}$$

Answer:

In order for a function to be a norm, it must satisfy positive homogeneity, definiteness, and the triangle inequality.

(a) $\|\alpha\mathbf{v}\| = |\alpha|\|\mathbf{v}\| \; \forall \, \alpha \in \mathbb{R}, \mathbf{v} \in \mathbb{R}^n$

$$\|\alpha\mathbf{v}\|_p := \left(\sum_{i=1}^{n} |\alpha v_i|^p\right)^{1/p} = \left(\sum_{i=1}^{n} |\alpha||v_i|^p\right)^{1/p} = |\alpha|^{p/p} \left(\sum_{i=1}^{n} |v_i|^p\right)^{1/p} = |\alpha|\|\mathbf{v}\|_p$$

(b) $\|\mathbf{v}\| = 0 \iff \mathbf{v} = 0$

$$\left(\sum_{i=1}^{n} |v_i|^p\right)^{1/p}$$

Case 1: $v_i < 0$, then $|v_i|^p > 0$
Case 2: $v_i > 0$, then $|v_i|^p > 0$ (Because of the absolute value)
Case 3: $v_i = 0$, then $|v_i|^p = 0$

In order for the sum to be zero, every element v_i has to be zero, so that $|v_i|^p = 0$.

(c) (Could not figure out, and this problem is not required)

Problem 4. (10 points)

Prove that $\lim_{p\to\infty} \|\mathbf{v}\|_p = \|v\|_\infty$

Answer: We define $\|\mathbf{v}\|_p := \left(\sum_{i=1}^{n} |v_i|^p\right)^{1/p}$

Let $\|\mathbf{v}\|_\infty = \max_i |v_i|$

$\|\mathbf{v}\|_p = \|\mathbf{v}\|_\infty \frac{\left(\sum_{i=1}^{n} |v_i|^p\right)^{1/p}}{\|\mathbf{v}\|_\infty}$

$= \|\mathbf{v}\|_\infty \left(\sum_{i=1}^{n} \frac{|v_i|^p}{\|\mathbf{v}\|_\infty^p}\right)^{1/p}$

$= \|\mathbf{v}\|_\infty \left(\sum_{i=1}^{n} \left(\frac{|v_i|}{\|\mathbf{v}\|_\infty}\right)^p\right)^{1/p}$

$\leq \|\mathbf{v}\|_\infty n^{1/p}$ (We know this because $\left(\frac{|v_i|}{\|\mathbf{v}\|_\infty}\right)^p \leq 1 \; \forall \, i \in \mathbb{N}_{\leq n}$ since $\|v\|_\infty \geq |v_i|$ for all i)

(*) We also know that $\|v\|_\infty \leq \|v\|_p \; \forall \, p \in \mathbb{N}_{<\infty}$

therefore, $\|v\|_\infty \leq \|v\|_p \leq \|v\|_\infty n^{1/p}$

Because of statement (*), we can say that $\|v\|_\infty \leq \lim_{p\to\infty} \|\mathbf{v}\|_p$

Therefore, $\|v\|_\infty \leq \lim_{p\to\infty} \|\mathbf{v}\|_p \leq \lim_{p\to\infty} n^{\frac{1}{p}} = \|v\|_\infty$

Since $\|v\|_\infty \leq \lim_{p\to\infty} \|\mathbf{v}\|_p \leq \|v\|_\infty$, we can conclude that $\lim_{p\to\infty} \|\mathbf{v}\|_p = \|v\|_\infty$,
as desired ■

Problem 5. (10 points)

Prove that $\|A\|_\infty = \max\limits_{1\le i\le n} \sum\limits_{j=1}^{m} |a_{ij}|$

Answer: $\|A\|_\infty = \max\limits_{\|\mathbf{v}\|_\infty=1} \|A\mathbf{v}\|_\infty$

$= \max\limits_{\|\mathbf{v}\|_\infty=1} \max\limits_{i=1,\dots,n} \left| \sum\limits_{j=1}^{m} a_{ij} v_j \right|$

$= \max\limits_{i=1,\dots,n} \max\limits_{\|\mathbf{v}\|_\infty=1} \left| \sum\limits_{j=1}^{m} a_{ij} v_j \right|$

$= \max\limits_{i=1,\dots,n} \sum\limits_{j=1}^{m} |a_{ij}|$ (We know this since $\|\mathbf{v}\|_\infty = 1$.)

$= \max\limits_{1\le i\le n} \sum\limits_{j=1}^{m} |a_{ij}|$, as desired ■

Problem 6. (10 points)

Suppose that $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$ and $\mathbf{x} \in \mathbb{R}^n$.
Let $f(\mathbf{x}) = \|\mathbf{b} - A\mathbf{x}\|_2^2$. Show that $\nabla f(\mathbf{x}) = 2A^\top(A\mathbf{x} - \mathbf{b})$

Answer: $f(\mathbf{x}) = \|\mathbf{b} - A\mathbf{x}\|_2^2$
Let $\mathbf{w} = \mathbf{b} - A\mathbf{x}$

$$\|\mathbf{w}\|_2^2 = \left(\left(\sum_{k=1}^{n} w_k^2\right)^{1/2}\right)^2 = \sum_{k=1}^{n} w_k^2 = \sum_{k=1}^{n} (\mathbf{b} - A\mathbf{x})_k^2$$

$$\nabla f(\mathbf{x}) = \nabla\|\mathbf{b} - A\mathbf{x}\|_2^2 = \frac{\partial}{\partial x_j} \sum_{k=1}^{n} (\mathbf{b} - A\mathbf{x})^2 = \sum_{k=1}^{n} \frac{\partial}{\partial x_j} (\mathbf{b} - A\mathbf{x})^2 = \sum_{k=1}^{n} 2\,(\mathbf{b} - A\mathbf{x})\,\frac{\partial}{\partial x_j}\,(\mathbf{b} - A\mathbf{x})$$

$$= \sum_{k=1}^{n} 2\,(\mathbf{b} - A\mathbf{x})\,\frac{\partial}{\partial x_j}\,(\mathbf{b} - A\mathbf{x})$$

(This is as far as I could get on number 6. I was just trying to manipulate the formulas but I did not know how to continue from here).

Attempt 2:

Problem 7. (10 points)

Let $A = \begin{bmatrix} 0 & 6 & -1 \\ 3 & 5 & 2 \\ 1 & -2 & 4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix}$

(a) Find L, U and P such that $PA = LU$ and L is lower triangular with ones on the diagonal, U is upper triangular and P is a permutation matrix.
(b) Use this LU decomposition to solve the linear system $A\mathbf{x} = \mathbf{b}$.

Answer:

(a) Let $PA^* = \begin{bmatrix} 3 & 5 & 2 \\ 0 & 6 & -1 \\ 1 & -2 & 4 \end{bmatrix}$ Where $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (Swap rows 1 and 2 of A), and $A^* = \begin{bmatrix} 3 & 5 & 2 \\ 0 & 6 & -1 \\ 1 & -2 & 4 \end{bmatrix}$. Let $L = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}$ and $U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$ such that $PA^* = LU$. $PA^*\mathbf{x} = \mathbf{b} \iff LU\mathbf{x} = \mathbf{b}$. Let $\mathbf{y} = U\mathbf{x}$. Then $L\mathbf{y} = \mathbf{b}$.

$$\begin{bmatrix} 3 & 5 & 2 \\ 0 & 6 & -1 \\ 1 & -2 & 4 \end{bmatrix} (R_3 - 1/3 \times R_1) = \begin{bmatrix} 3 & 5 & 2 \\ 0 & 6 & -1 \\ 0 & -11/3 & 10/3 \end{bmatrix} (R_3 + 11/18 \times R_2)$$

$$= \begin{bmatrix} 3 & 5 & 2 \\ 0 & 6 & -1 \\ 0 & 0 & 49/18 \end{bmatrix} = U. \; L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1/3 & 11/18 & 1 \end{bmatrix} \text{ (Inserting the } k \text{ values).}$$

Finally, we have found L, U, and P.

(b) $PA^*\mathbf{x} = \mathbf{b} \iff LU\mathbf{x} = \mathbf{b}$. Let $\mathbf{y} = U\mathbf{x}$. Then $L\mathbf{y} = \mathbf{b}$.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1/3 & 11/18 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix}$$

$y_1 = 1$.
$0y_1 + y_2 = 5$. $y_2 = 5$.
$-1/3y_1 + 11/18y_2 + 1y_3 = 3$. $-1/3 + 55/18 + y_3 = 3$. $y_3 = 5/18$.

$\mathbf{y} = \begin{bmatrix} 1 \\ 5 \\ 5/18 \end{bmatrix}$. $U\mathbf{x} = \mathbf{y}$. $\begin{bmatrix} 3 & 5 & 2 \\ 0 & 6 & -1 \\ 0 & 0 & 49/18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 5/18 \end{bmatrix}$

$49/18x_3 = 5/18$. $x_3 = 5/49$
$6x_2 - 1x_3 = 5$. $6x_2 - 5/49 = 5$. $6x_2 = 250/49$. $x_2 = 250/294 = 125/147$.
$3x_1 + 5x_2 + 2x_3 = 1$. $3x_1 + 5(125/147) + 2(5/49) = 1$. $3x_1 + 625/147 + 10/49 = 1$.
$3x_1 = -508/147$. $x_1 = -508/441$.

$$\mathbf{x} = \begin{bmatrix} -508/441 \\ 125/147 \\ 5/49 \end{bmatrix}$$

Problem 8. (10 points)

Use the normal equations to solve the least squares problem $x^* = \text{argmin}\|\mathbf{b} - A\mathbf{x}\|_2$ where $A = \begin{bmatrix} 1 & 5 \\ -3 & 2 \\ 4 & -7 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 4 \\ -3 \\ 2 \end{bmatrix}$

Answer: Through derivation, we know that the normal equation is as follows:
$\mathbf{x}^* = \left(A^T A\right)^{-1} A^T \mathbf{b}$

$$A^T = \begin{bmatrix} 1 & -3 & 4 \\ 5 & 2 & -7 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & -3 & 4 \\ 5 & 2 & -7 \end{bmatrix} \begin{bmatrix} 1 & 5 \\ -3 & 2 \\ 4 & -7 \end{bmatrix} = \begin{bmatrix} 26 & -29 \\ -29 & 78 \end{bmatrix}$$

$$\left(A^T A\right)^{-1} = \begin{bmatrix} 26 & -29 \\ -29 & 78 \end{bmatrix}^{-1} = \begin{bmatrix} 78/1187 & 29/1187 \\ 29/1187 & 26/1187 \end{bmatrix}$$

$$\left(A^T A\right)^{-1} A^T = \begin{bmatrix} 78/1187 & 29/1187 \\ 29/1187 & 26/1187 \end{bmatrix} \begin{bmatrix} 1 & -3 & 4 \\ 5 & 2 & -7 \end{bmatrix} = \begin{bmatrix} 223/1187 & -176/1187 & 109/1187 \\ 159/1187 & -35/1187 & -66/1187 \end{bmatrix}$$

$$\left(A^T A\right)^{-1} A^T \mathbf{b} = \begin{bmatrix} 223/1187 & -176/1187 & 109/1187 \\ 159/1187 & -35/1187 & -66/1187 \end{bmatrix} \begin{bmatrix} 4 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1638/1187 \\ 609/1187 \end{bmatrix} = \mathbf{x}^*$$